Question ID: 4b99b481

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Medium

Question

While researching a topic, a student has taken the following notes:

- Scientists have developed a “freeze-thaw” battery that can retain 92% of its charge after twelve weeks.
- The battery contains molten salt (a type of salt that liquifies when heated and solidifies at room temperature).
- When the salt is in a liquid state, energy flows through the battery.
- When the salt is in a solid state, energy stops flowing and is stored in the battery.
- The stored (frozen) energy can be used by reheating (thawing) the battery.

The student wants to specify how the salt enables energy storage. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Scientists have developed a freeze-thaw battery that contains molten salt, which liquifies when heated and solidifies at room temperature.

B. The stored energy in a freeze-thaw battery, which contains molten salt, can be used by reheating the battery.

C. When the molten salt in a freeze-thaw battery solidifies at room temperature, energy stops flowing and can be stored in the battery.

D. Molten salt allows a freeze-thaw battery to retain 92% of its charge after twelve weeks.

Correct Answer: C

Rationale

Choice C is the best answer. The sentence specifies how the salt in a freeze-thaw battery enables energy storage, explaining that energy stops flowing and can be stored when the salt solidifies at room temperature.

Choice A is incorrect. The sentence explains some properties of molten salt; it doesn’t specify how that salt enables energy storage. Choice B is incorrect. The sentence indicates how the energy in a freeze-thaw battery can be released; it doesn’t specify how the salt in the battery enables energy storage. Choice D is incorrect. The sentence specifies how much charge the freeze-thaw battery retains when storing energy; it doesn’t specify how the salt in the battery enables energy storage.